

1 • Shared evidence: orchard records

The flower-to-seed investigation

Name _____ Date _____

These are constructed observation scenarios, not results from a class experiment.

A • Transfer event

A bee carries pollen from an anther to a stigma of a suitable flower. This is pollen transfer, not the fusion event.

C • Development

After successful fertilisation, an ovule develops into a seed; the ovary develops into fruit. Other conditions also affect success.

E • Incomplete evidence

Pollen is seen on a stigma. The inside of the ovule has not been observed. Pollen transfer alone does not prove fertilisation occurred.

B • Inside the flower

After compatible pollen arrives, a pollen tube can grow. A male gamete can reach an egg cell in an ovule and fuse with it.

D • Wrong destination

Pollen is seen on a leaf. No pollen is seen on the stigma in this record. The record does not show pollination.

F • Model boundary

This model follows sexual reproduction in a flowering plant. It omits detailed cell structures and time scales; no class growing experiment took place.

W1. Compare records A and D. Which shows pollination?

W2. Order A, B and C and name the three processes.

W3. What can record E establish, and what remains unobserved?

2 • Supported route: follow the pollen

Word bank: anther / stigma / ovule / pollination / fertilisation.

S1. Complete the transfer: pollen moves from an ____ to a ____ .

S2. What is that transfer called?

A. Pollination

B. Fertilisation

S3. Complete: The fusion of a male gamete and an egg cell is ____ .

S4. Which develops into a seed?

A. A fertilised ovule

B. A pollen grain on a leaf

S5. Order: seed develops / pollen reaches stigma / gametes fuse. Explain one arrow.

3 • Standard route: process investigator

Use source cards A-F.

Process	What happens?
Pollination	
Fertilisation	
Seed formation	

N1. Complete the process table.

N2. Trace pollen's role from anther to a male gamete reaching an egg cell. Name the ovule.

N3. Explain why pollen on a leaf is not evidence of pollination.

N4. Record E shows pollen on a stigma. Explain what is known and what cannot yet be claimed.

4 • Stretch route: audit the orchard claim

Use precise biological structures and bounded evidence.

T1. Explain the full sequence from pollen transfer to seed formation. Include anther, stigma, gamete and ovule.

T2. Repair “Bees carry seeds to a flower, then pollen becomes the fruit”.

T3. Does observing pollen on a stigma guarantee a mature seed? Explain two later requirements or uncertainties.

T4. Design a record that separates observation of pollen arrival from later seed development. What would each observation establish?

5 • Exit and reflection

Use a short explanation or labelled route.

E1. What is pollination?

E2. How is fertilisation different?

E3. What develops into a seed, and what develops into fruit?

E4. Why does pollen on a leaf not establish pollination?

Reflection. Which later stage would you investigate next?
